Advance Python Assignment

Topic-10: Search and Match Functions

Q1. Explain re.search() function in Python.

Ans.
The re.search() function in Python is used to search for a pattern in a string. It scans the entire string and returns the first match if the pattern is found. If the pattern is not found, it returns None. This function is part of the re module which is used for working with regular expressions. It is useful when we want to check whether a specific word or pattern exists anywhere in a string.

Example:
```
import re

text = "Hello Anshu, welcome to Python"
result = re.search("Anshu", text)

if result:
   print("Match found")
else:
   print("Match not found")
```

Q2. Explain re.match() function in Python.

Ans.
The re.match() function in Python is used to match a pattern only at the beginning of a string. If the pattern is found at the start, it returns a match object, otherwise it returns None. It is also part of the re module and is used for pattern matching. Unlike re.search(), it does not scan the whole string, but checks only from the beginning. This makes it useful when we want to validate starting patterns.

Example:
```
import re

text = "Hello Anshu"
result = re.match("Hello", text)
```

```
if result:
    print("Match found at beginning")
else:
    print("No match")
```